

**PT. INDONESIA POWER
UNIT JASA PEMBANGKITAN PLTU JERANJANG
INTEGRATED MANAGEMENT SYSTEM**

21.TE/CBM/UJPJRJ/2020

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Equipment : CWP 1 Unit 3

KKS : LJ30PAC10AP001-001

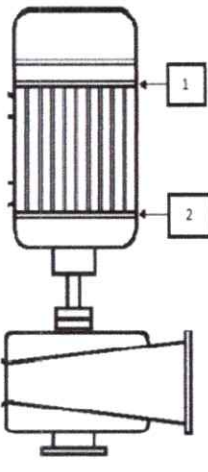
Technology : Vibrasi dan IR Thermography

Tanggal : 04 Juni 2020

Status : Warning

Finding : Vibrasi CWP 1 Unit 3 mengalami kenaikan level severity

I. Data Spesifikasi Motor

MOTOR		POMPA	
Type, Mfg	: YKKL-450-6TH	Type, Mfg.	:
Speed	: 989	Speed	:
Power	: 355	Bearing Type	:
Bearing Type	: Roller Bearing	Inboard Bearing	:
Inboard Bearing	: 6226	Onboard Bearing	:
Outboard Bearing	: 7330B	Total Blade	:
Rotor Bar	: -		
Foundation	: Rigid		
FOTO EQUIPMENT		POINT PENGUKURAN	
			

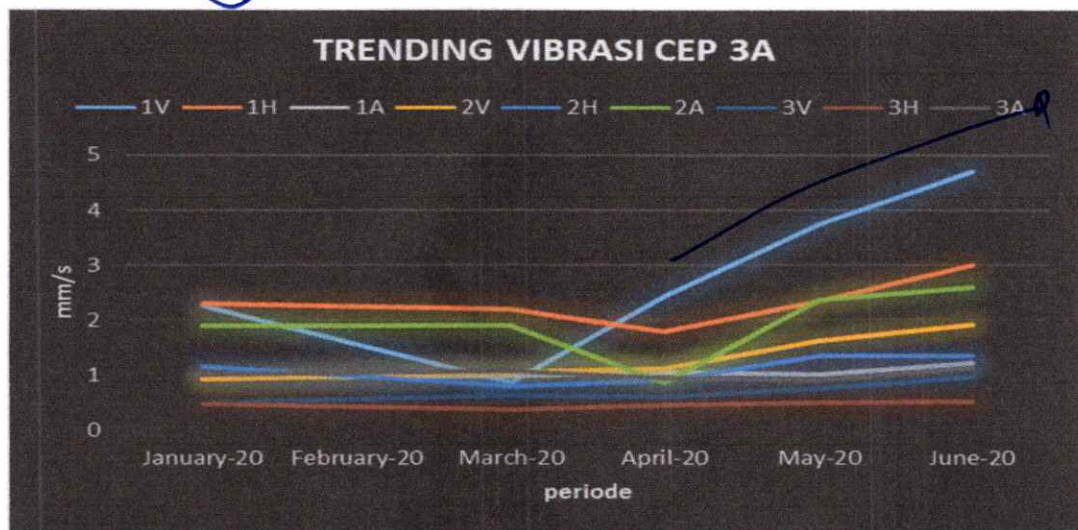
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II. Hasil Pengukuran Vibrasi

BULAN	CWP 1 UNIT 3								
	MOTOR						POMPA		
	1V	1H	1A	2V	2H	2A	3V	3H	3A
January-20	2.26	2.29	0.94	0.92	1.14	1.9	0.49	0.47	1.01
March-20	0.85	2.19	0.97	1.04	0.79	1.89	0.62	0.37	1.07
April-20	2.42	1.79	1.05	1.11	0.89	0.85	0.59	0.45	0.94
May-20	3.75	2.34	0.99	1.62	1.35	2.37	0.74	0.48	0.89
June-20	4.71	2.99	1.21	1.91	1.34	2.59	0.96	0.51	1.13



STANDART ISO 10816-3 (GROUP 1 RIGID)

	≥0	≥2.3	≥4.5	≥7.1
ALARM LIMIT	A	B	C	D

✓

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Hasil Rekaman Spectrum Vibrasi :

✓ **Pada Motor :**

- Pada Bearing 1 (NDE Motor) terjadi vibrasi alarm dengan nilai 4.71 mm/s RMS, indikasi mulai structure looseness

✓ **Pada Pompa :**

- Nilai overall Vibrasi masih rendah status tertinggi normal

Rekomendasi :

✓ **Pada Motor :**

- Perencanaan Change over CWP
- Cleaning Intake
- Pengecekan Kekencangan baut coupling
- Pengecekan kekencangan baut koneksi antara motor dan pompa

✓ **Pada Pompa :**

- Lakukan monitoring vibrasi secara rutin

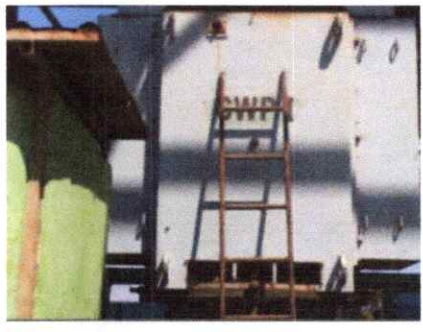
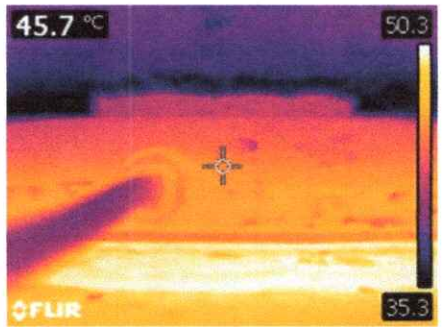
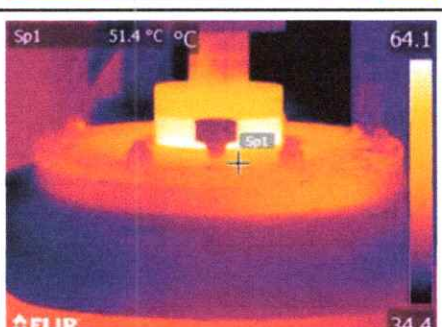
III. Hasil Pengukuran IR Thermography

No. KKS	Description	Measurement Point	Upper Warning Limit Maximo	Upper Action Limit Maximo	Analysis Standard	Status per Point Measurement	Overall Status Equipment	T max Reference	Data Analysis	
									Date	(°C)
LJ30PAC10AP001-001	EQUIPMENT-CIRCULATING WATER PUMP A UNIT 3	Cover winding motor	71	81	FLIR	Low	Low	60	4-Jun-20	65.3
		Bearing 1 motor	61	71	FLIR	Normal		50	4-Jun-20	45.7
		Bearing 2 motor	61	71	FLIR	Normal		50	4-Jun-20	47.7
		Bearing 3 pump	71	81	FLIR	Normal		60	4-Jun-20	51.4
		Connection Phasa R	41	51	FLIR	Normal		30	4-Jun-20	23.7
		Connection Phasa S	41	51	FLIR	Normal		30	4-Jun-20	23.8
		Connection Phasa T	41	51	FLIR	Normal		30	4-Jun-20	23.6

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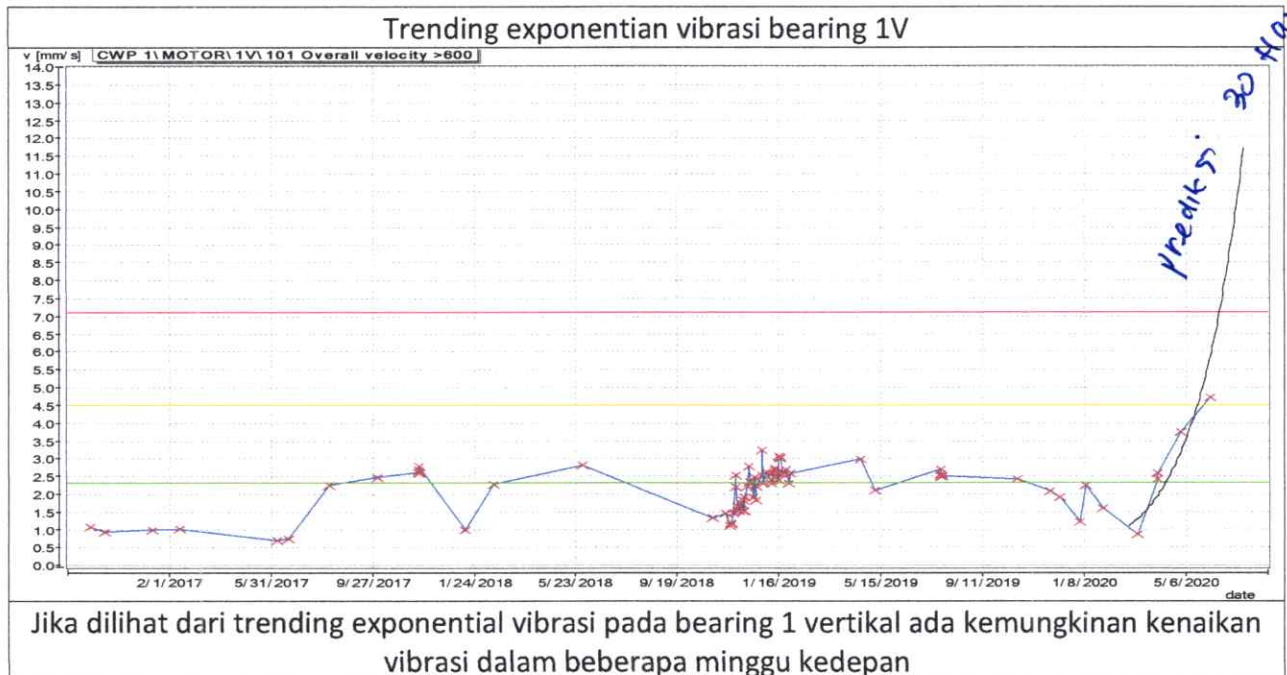
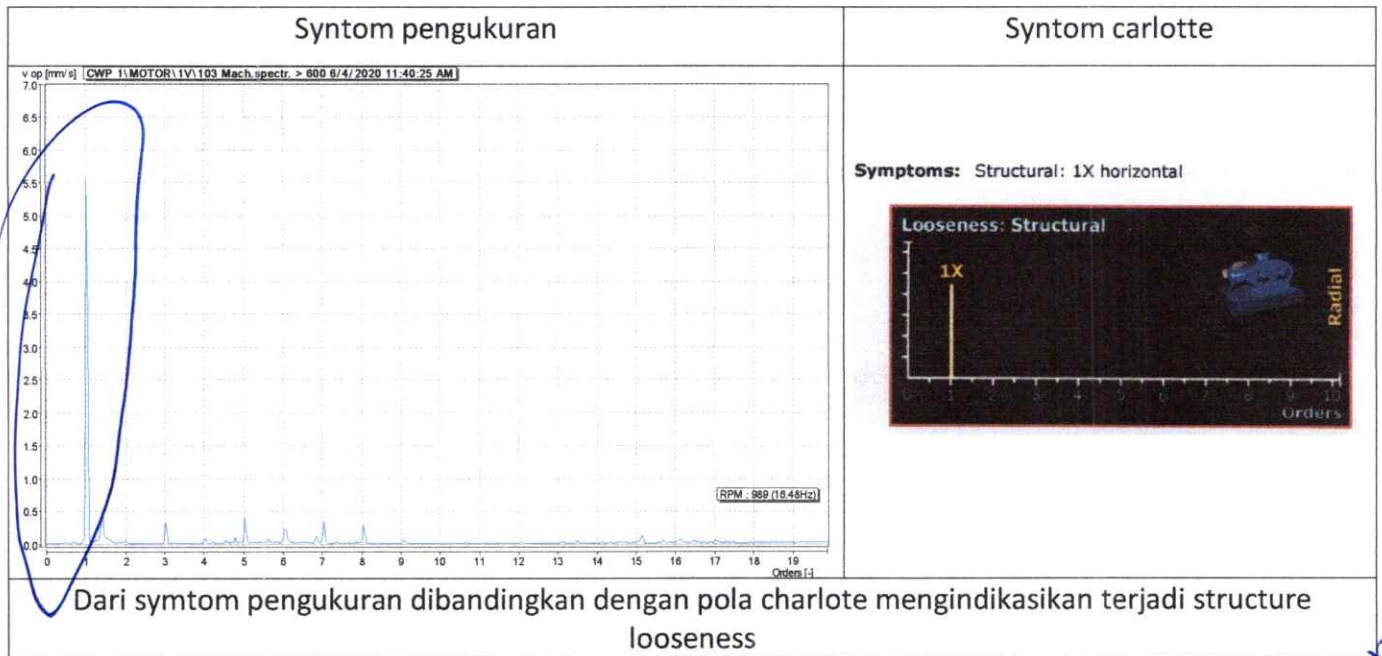
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CITRA IMAGING INFRARED THERMOGRAPHY	FOTO EQUIPMENT ACTUAL	KETERANGAN
		Pada body / cover winding CIRCULATING WATER PUMP 3 A temp tertinggi 65.3 degC
		Bearing 1 motor CIRCULATING WATER PUMP 3 A temp tertinggi 45.7 degC
		Bearing 2 motor CIRCULATING WATER PUMP 3 A temp tertinggi 47.7 degC
		Bearing 1 motor CIRCULATING WATER PUMP 3 A temp tertinggi 51.4 degC

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IV. Analisa :



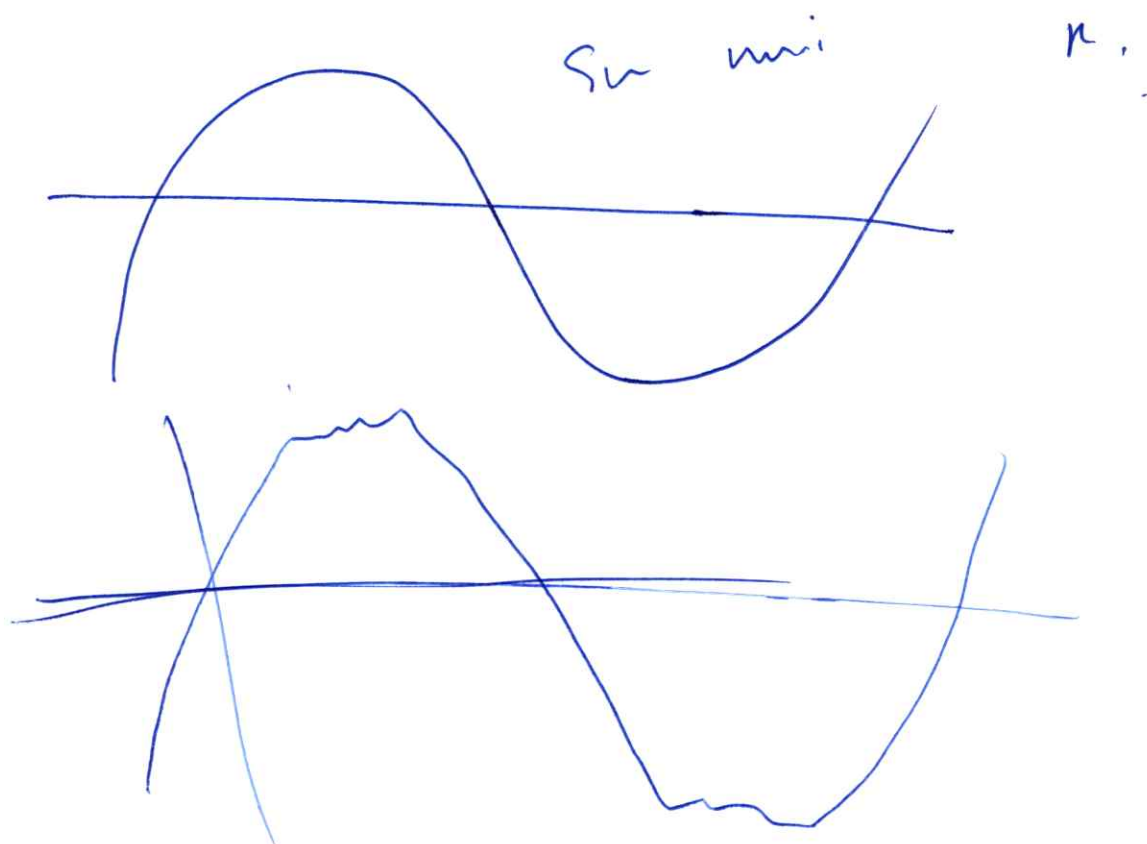
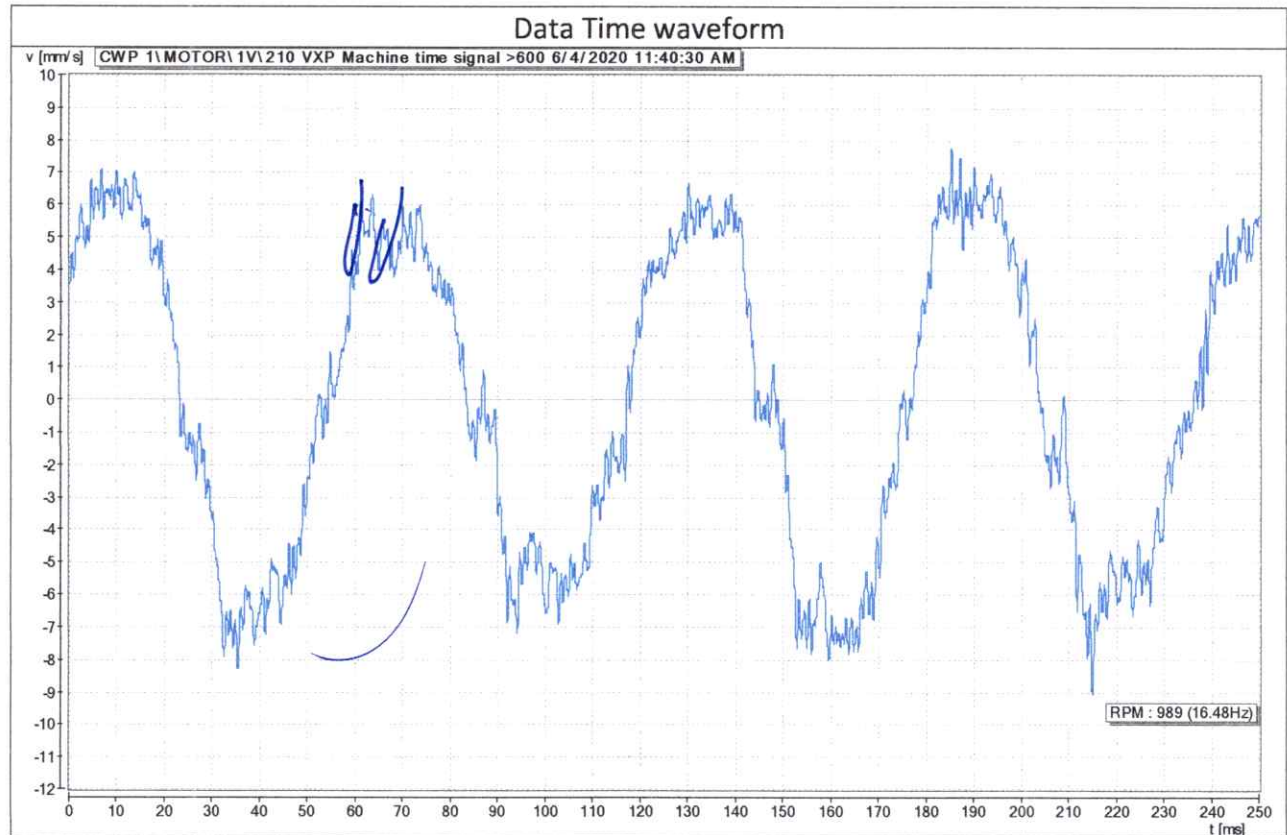
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IV. Kesimpulan :

- Terjadi kenaikan vibrasi di bearing 1 arah vertical

V. Rekomendasi :

No	Rekomendasi	PIC	Freq	Time line
1.	Persiapan Change Over CWP 3A ke 3B	Operasi	1 kali	5-19 Juni 2020
2.	Cleaning Intake	Pemeliharaan	1 kali	5-19 Juni 2020
3.	Pengecekan kekencangan baut coupling	Pemeliharaan	1 kali	5-19 Juni 2020
4.	Pengecekan kekencangan baut koneksi motor dan pompa	Pemeliharaan	1 kali	5-19 Juni 2020
5.	Pemantauan Vibrasi	PdM	1 per minggu	5-19 Juni 2020

6. Pemantauan parameter cur ^{up} 2x dari normal 5-19 Juni

JERANJANG, 04 JUNI 2020

Mengetahui

Ricky Rinaldi

Dibuat Oleh

Hermawan